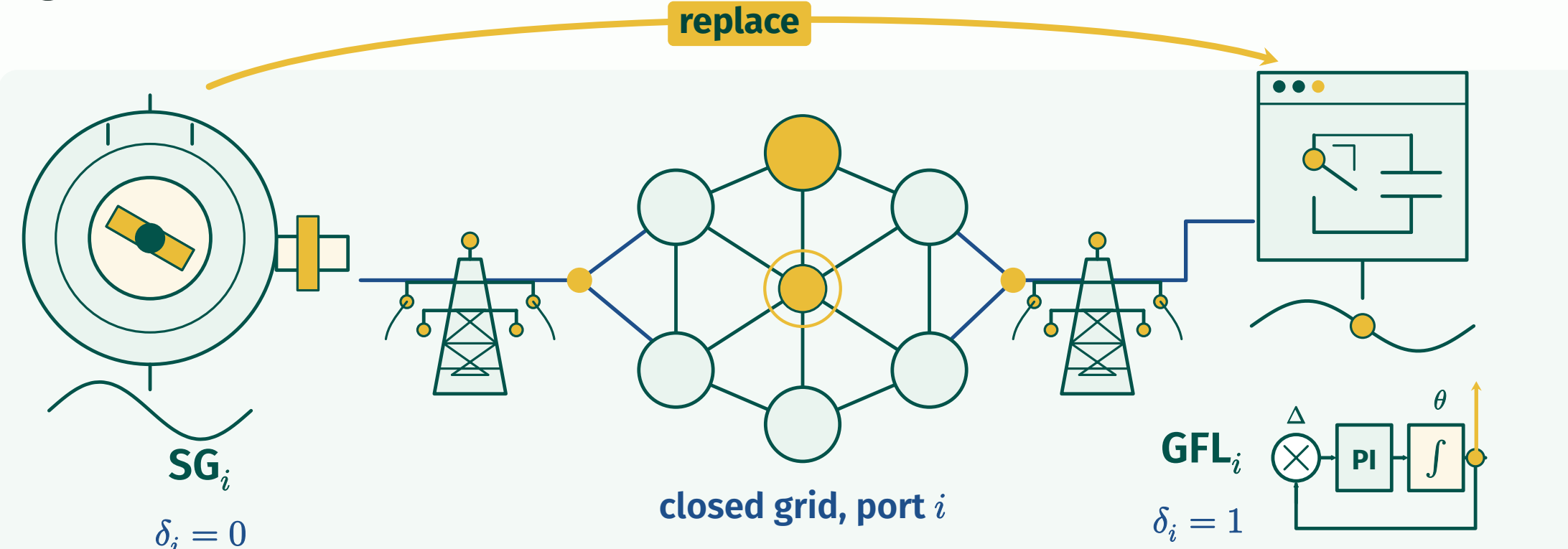


1 THE PROBLEM: SAFE SG → GFL TRANSITION

A replacement is local hardware, but stability is a property of the closed grid and its controller architecture.



Safe SG → GFL replacement is a joint portfolio and controller design problem.

Engineering objective $S(\delta) = \{i : \delta_i = 1\}$
 $\max_{S, K} \sum_{i \in S} P_i^0$ s.t. $\alpha_{\perp}(S, K) \leq -\sigma_{\text{req}}$, $K \in \mathcal{K}_n$
 P_i^0 SG_i dispatch $\alpha_{\perp}$ rightmost non-gauge root σ_{req} decay margin $\mathcal{K}_n$ gains using $\leq n$ hops

- Choose portfolio δ
- Choose radius n : $K \in \mathcal{K}_n$
- Design gains K_p, K_I
- Verify grid spectrum, events

Report the largest independently verified portfolio found.
Certified global optimality is future work.

Refs: [1] [5] [6]

2 MODEL: DAE → TRANSVERSE STABILITY

$E(x, \delta) \dot{x} = f(x, z; \delta, K, \tau)$
 $0 = g(x, z; \delta)$
 x states, z network, δ portfolio, K gains, τ delays
equilibrium: $f(x^*, z^*) = 0$, $g(x^*, z^*) = 0$

$$\begin{bmatrix} \Delta \dot{x} \\ 0 \end{bmatrix} = \begin{bmatrix} f_x & f_z \\ g_x & g_z \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta z \end{bmatrix}$$

$$A = f_x - f_z g_z^{-1} g_x$$

Schur complement, g_z invertible

$$A_{\perp} = Z^T A Z$$

Z : orthonormal basis of $\gamma^{\perp}$ (gauge $A\gamma = 0$)

$$\alpha_{\perp}(S, K) = \max \text{Re } \sigma(A_{\perp})$$

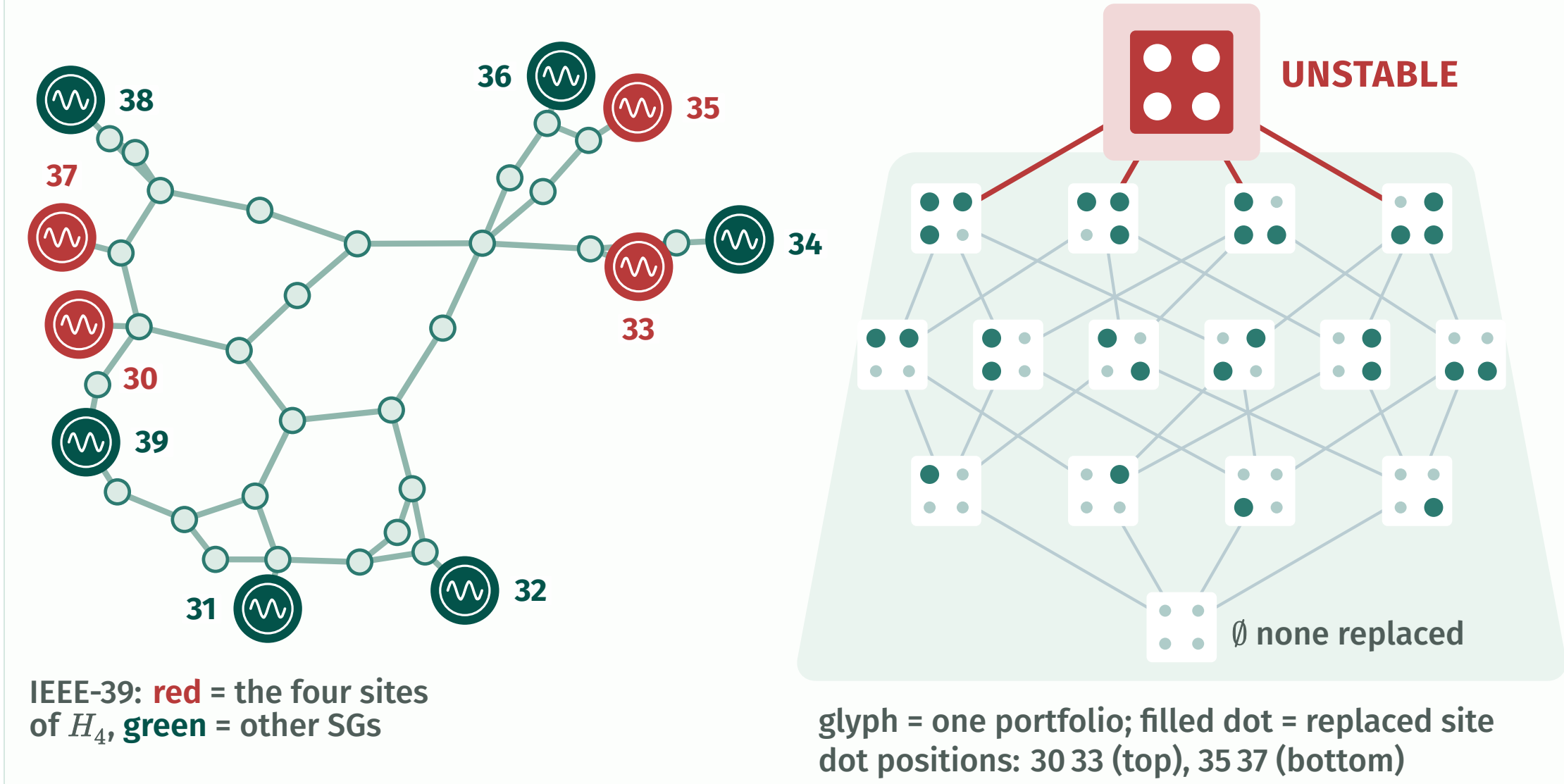
$\alpha_{\perp} < 0$: stable
 $\alpha_{\perp} > 0$: unstable

With PLL delay, $\alpha_{\perp}$ is the rightmost root of $\det \Delta_{\perp}(s)$;
 $\Delta(s) = sI - A_0 - B E(s) C$, exact exponentials.

Refs: [1] [7]

3 IEEE-39: MINIMAL COLLECTIVE BLOCKER

SEPARATE TX4 FIXED-POLICY MODEL CONTRACT (NO PLL DELAY)



Minimal collective blocker $H_4 = \{30, 33, 35, 37\}$

$$\alpha_{\perp}(R) < 0 \quad \forall R \subsetneq H_4, \quad \text{but} \quad \alpha_{\perp}(H_4) = +0.127 \text{ s}^{-1} > 0$$

Every proper subset stays stable (15 of 15); adding the fourth site pushes one mode at $f = 0.6223$ Hz into the right half-plane.

Canonical nominal witness, not a universal blocker.

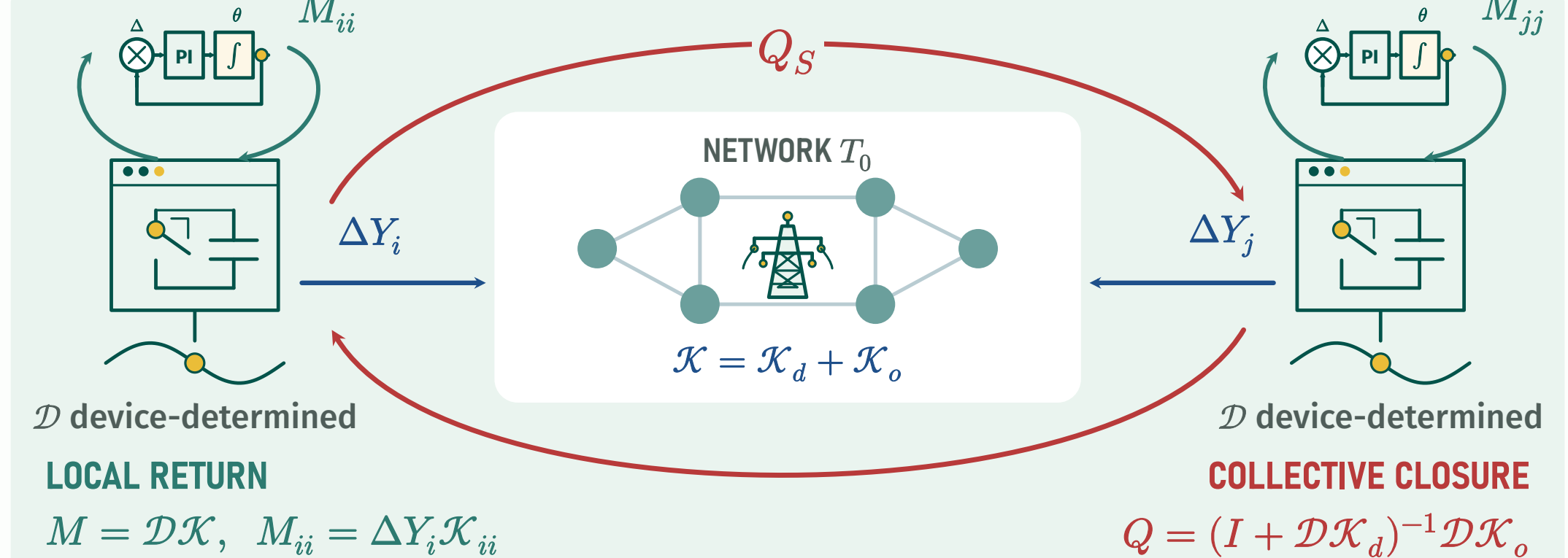
Refs: [5]

4 EXACT NETWORK CLOSURE: LOCAL × COLLECTIVE

$$T_S(s) = T_0(s) + \sum_{i \in S} E_i \Delta Y_i(s) E_i^T$$

$$\Delta Y_i = E_i^T [T_{\{ij\}}(s) - T_0(s)] E_i, \quad \mathcal{K} = E^T T_0^{-1} E = \mathcal{K}_d + \mathcal{K}_o$$

A Two devices, one network



$$\det(I + M_S) = \prod_{i \in S} \det(I + M_{ii}) \cdot \det(I + Q_S)$$

Audit: stored boundary $H_4 = \{30, 33, 35, 37\}$, $f \approx 0.7064$ Hz
 $\min_i \sigma_{\min}(I + M_{ii}) = 0.4892$, $\sigma_{\min}(I + Q_{H_4}) = 4.39 \times 10^{-8}$
Delayed full IEEE-39, 44 ms: $\text{eig}(Q) = -1$ at every unstable root, all $|L_i| \geq 0.011$.
All local factors stay regular; the instability enters through $\det(I + Q_S)$.

Refs: [7] [8]

5 BEYOND NODAL DAMPING: EFFECTIVE OPERATORS

A Exact Schur self-energy

$$S_r(s) = sI - A_{rr} - A_{rc}(sI - A_{cc})^{-1} A_{cr}$$

$$T(s) = s^2 M + s D_0 + L + \Pi(s)$$

$$L_{\text{eff}} = L + \Pi_e(0), \quad D_{\text{eff}} = D_0 + \Pi_o(s)/s$$

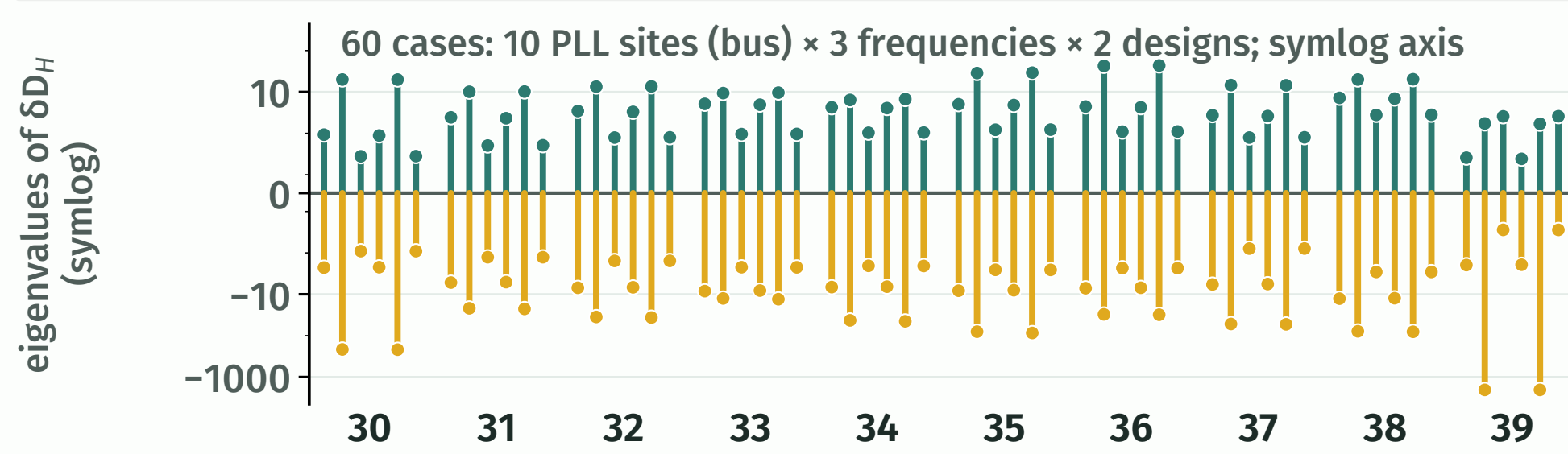
$$M_{\text{eff}} = M + [\Pi_e(s) - \Pi_e(0)]/s^2$$

$$T = s^2 M_{\text{eff}} + s D_{\text{eff}} + L_{\text{eff}}$$

C Damping operator $D_H(\Omega) = \text{Herm } Z(j\Omega)$

$$\delta D_H = \frac{1}{2}(a v^H + v a^H)$$

one positive, one negative eigenvalue



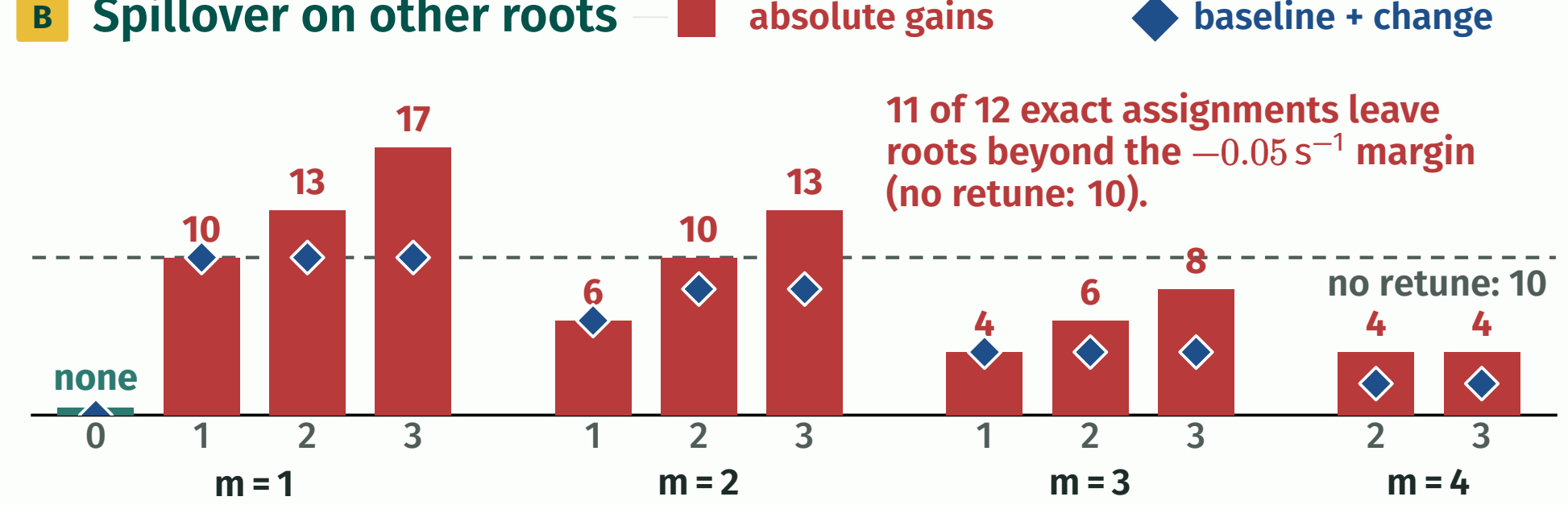
damping ↑ One local change raises damping along one direction and lowers it along another; damping at a single frequency is not a stability test.
damping ↓

Refs: [8]

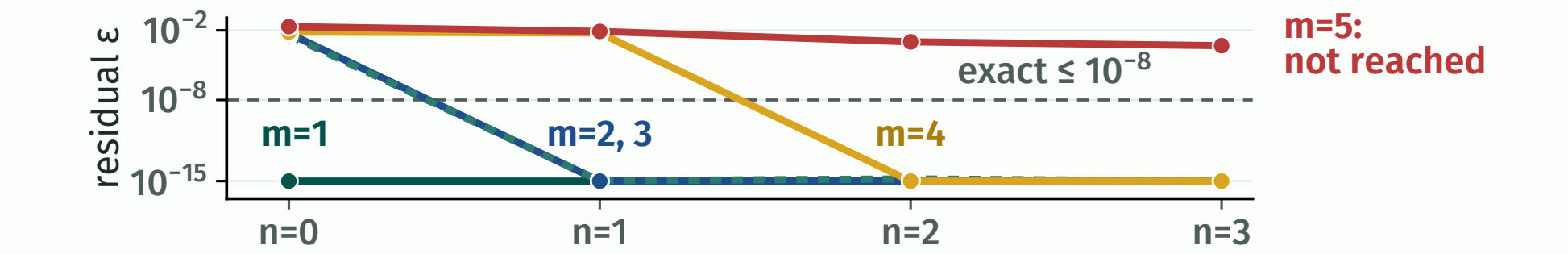
8 N-HOP RESULTS: AUTHORITY, SPILLOVER, REPAIR

Full IEEE-39, exact delay 44 ms

B Spillover on other roots absolute gains baseline + change



A Authority: residual vs radius



Each extra mode pinned needs more measured neighbors ($2m \leq 2|\mathcal{N}_i|$).

Spectral repair at 44–52 ms already works with $n = 0$; the clean-spectrum share is 90% for every n (events fail there, panel 9) and 92.5% fails. Declared sweep, not an optimum.

Exact assignment of selected modes does not guarantee full-spectrum stability.

Refs: [2]

6 GRID-EMBEDDED PLL GAIN SYNTHESIS

Open the PLL ports, target $\lambda = \alpha + j\omega$

$$Az = B\theta, \quad e = Cz + D\theta \Rightarrow G = D + CA^{-1}B$$

Row condition: $y = G(\lambda; \rho) q$

$$(\lambda K_{p,i} + K_{I,i}) e^{-\lambda \tau_i} y_i = \lambda^2 (1 + t_{f,i} \lambda) q_i$$

$$\text{from } s\theta = \omega, \quad t_{f,i} s\omega = \xi + K_p e_d - \omega, \quad s\xi = K_I e_d$$

$$W_i = \frac{\lambda^2 (1 + t_{f,i} \lambda) e^{\lambda \tau_i} q_i}{y_i} \quad W_i = \lambda K_{p,i} + K_{I,i}$$

$$K_{p,i} = \frac{\text{Im } W_i}{\omega}, \quad K_{I,i} = \text{Re } W_i - \alpha K_{p,i}$$

$$K = \Phi(\rho, \lambda, q, \tau)$$

The gain is local; its design is embedded in the grid mode.

$$[\lambda K_p(h) + K_I(h)] e^{-\lambda(\tau+h)} = [\lambda K_p + K_I] e^{-\lambda \tau}$$

$$\left. \frac{d\mu}{dh} \right|_0 = -K_p(\mu - \lambda)(\mu - \bar{\lambda}) \frac{\partial \mu}{\partial K_I}$$

Preserving one mode exactly does not preserve the others.

Full IEEE-39, exact delay 44 ms

no retune
8 roots
events abort

protect 4.91 Hz
0 roots
2/5 events pass

NUMERICAL
protect 1.04 Hz
0 roots
5/5 events pass

abort = integrator stopped

Refs: [3] [6]

9 DELAYS, CO-DESIGN, AND NONLINEAR CHECKS

A Unequal latencies mix graph modes

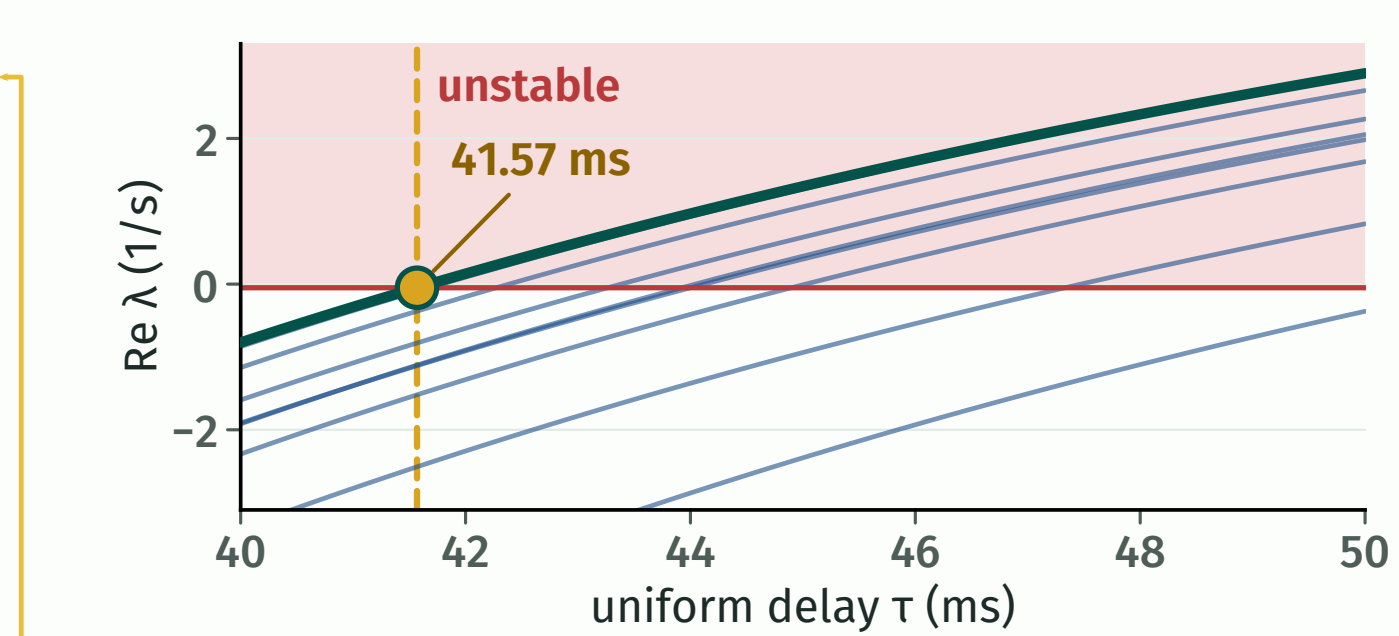
$$E_r(s) = \text{diag } e^{-s\tau_i}, \quad [L, K(L)E_r] = K(L)[L, E_r]$$
$$\| [L, E_r(j\Omega)] \|_F^2 = 8 \sum_{ij \in \mathcal{E}} w_{ij}^2 \sin^2 \left(\frac{\Omega(\tau_i - \tau_j)}{2} \right)$$

zero only if $\tau_i = \tau_j$ on every edge

C Co-design loop

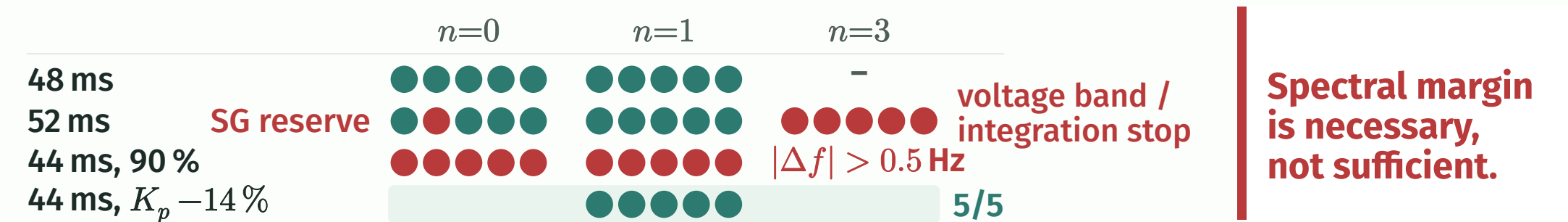
- Choose portfolio
- Equilibrium, spectrum
- Closure Q, D_H
- Radius n
- Gain map, predictor
- FULL-SPECTRUM corrector
- Nonlinear events

B Delay root locus



gold marker: first crossing of the -0.05 s^{-1} margin; full IEEE-39, exact delay

D Five frozen nonlinear events



Refs: [4] [9]

MAIN CONTRIBUTIONS

- Minimal replacement blockers:** the H_4 witness is unstable although all its proper subsets are stable.
- Exact closure factorization** of the network return into a local and a collective part.
- Beyond nodal damping:** self-energy and signed damping redistribution.
- Grid-embedded PLL gain synthesis** and exact delay transport.
- Information radius** on the full IEEE-39: authority grows, safety must still be verified.

KEY RESULTS

model	finding	evidence
TX4 IEEE-39	H_4 blocker, 15/15 proper subsets stable, $\alpha_{\perp}(H_4) = +0.127 \text{ s}^{-1}$	full-spectrum, fixed policy
full IEEE-39, exact delay	first margin crossing 41.57 ms; uniform $K_p - 14\%$ restores margin, 5/5 events	argument-principle counts + Julia events
n -hop, full IEEE-39	$n_{\min} = 0, 1, 1, 2$ for $m = 1-4$; 11/12 exact assignments leave roots beyond the margin	modal equations + full counts
δD_H	60/60 cases with one positive and one negative eigenvalue	Julia validation

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